1 => CREATION OF IMAGE MAP TO FIX HOTSPOTS

Program:

main.html

```
<html>
<head>
<title>India Map</title>
</head>

<body>
<img src="https://www.mapsofindia.com/images2/india-map-2019.jpg"
usemap="#indiamap">

<map name="indiamap">
    <area shape="circle" coords="340,971,107"
    href="tamilnadu.html" alt="Tamilnadu">

    <area shape="rect" coords="247,827,354,1007"
    href="karnataka.html" alt="Karnataka">
</map>

</body>
</html>
```

Karnataka.html

```
<!DOCTYPE html>
<html>
<head>
<title>Karnataka - India</title>
</head>

<body style="background-color:wheat;">

<h1 style="text-align:center;">Karnataka</h1>

<p>
Karnataka is a southern Indian state known for its tech capital,
Bengaluru. It has a rich history with UNESCO sites like Hampi
and beautiful landscapes.
</p>

<ul>
<li>Districts: 30</li>
<li>Capital City: Bangalore</li>
<li>Largest City: Bangalore</li>
<li>Governor:  Dr. Sudha Yadav</li>
<li>Chief Minister:  D. K. Shivakumar</li>
<li>Population: 69,074,000</li>
<li>Tourist Spots: Gol Gumbaz, Mysore Palace, Keshava Temple</li>
</ul>

<a href="main.html">Back</a>

</body>
</html>
```

Tamilnadu.html

```
<!DOCTYPE html>
<html>
<head>
<title>Tamil Nadu - India</title>
</head>

<body style="background-color:palegreen;">

<h1 style="text-align:center;">Tamil Nadu</h1>

<p>
Tamil Nadu is a southern Indian state. Its capital and largest
city is Chennai. It is bordered by Puducherry, Kerala,
Karnataka and Andhra Pradesh.
</p>

<ul>
<li>Districts: 38</li>
<li>Capital City: Chennai</li>
<li>Largest City: Chennai</li>
<li>Governor: Rajendra Arlekar</li>
<li>Chief Minister: C. Joseph Vijay</li>
<li>Population: 77,582,000</li>
<li>Tourist Spots: Mamallapuram, Ooty, Kodaikanal,
Marina Beach, Madurai Meenakshi Temple, Thanjavur</li>
</ul>

<a href="main.html">Back</a>

</body>
</html>
```

2 => Creation of Web Page Using All Types of Cascading Style Sheets

Program:

Style.html

```
<!DOCTYPE html>
<html>
<head>
<title>FLOWERS</title>

<!-- External CSS -->
<link rel="stylesheet" href="style.css">

<!-- Embedded CSS -->
<style>
p {
    background-color: lightgrey;
    text-align: justify;
    margin: 2em 7em;
}
</style>
</head>

<body>
<h1>FLOWER</h1>

<p>
<span style="font-weight:200; font-size:x-large; font-
family:fantasy;">
Flower
</span>
sometimes known as a bloom or blossom, is the reproductive structure
found in flowering plants.
The flower is God's finest workmanship in the world. It is his finest
gift to mankind.
```

```
We have seen flowers of many kinds and many colors. In India we see
flowers like:
</p>

<!-- Inline CSS -->
<table style="margin:auto; text-align:center; padding:3px;">
<tr>
<td align="left">
<div class="div">
<ul>
<li><a href="https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcSTgtoKchwB7lfBAKx_U4cGrziXuzWn31Ue
lQ&s">Lily</a></li>
<li><a href="https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcTqQi7iFH1j3qpXA4lxTPSaxI0NAX9PXNUy
Tw&s">Lotus</a></li>
<li><a href="https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcQCsweBYOj8Z_aIeivnhuxdsw_OUthYAsUT
NQ&s">Rose</a></li>
<li><a href="https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcS0QpzKsDSXBxzcjQ5eh_7yuG6tDgedgM_i
w&s">Jasmine</a></li>
</ul>
</div>
</td>
</tr>
</table>

</body>
</html>
```

Style.css

```
h1, h2 {
    text-decoration: underline;
    font-style: italic;
    text-align: center;
}

#body {
    background-color: tan;
    border: 2px dotted red;
    text-align: center;
}

.div {
    border: 2px solid peru;
    padding: 10px;
}

* {
    letter-spacing: 1px;
}

a:link { color: black; }
a:visited { color: yellow; }
a:hover { color: green; }
a:active { color: blue; }

ul li {
    font-size: small;
}
```

3 => Client-Side Validation of Web Form Controls Using DHTML

Program:

Control.html

```
<!DOCTYPE html>
<html>
<head>
<title>Form Validation</title>

<script>
function validateForm() {
    var x = document.forms["myForm"]["fname"].value;

    if (x == "") {
        alert("Name must be filled out");
        return false;
    }
}
</script>

</head>
<body>

<form name="myForm" action="wlcm.html"
onsubmit="return validateForm()" method="post">

Name: <input type="text" name="fname">
<input type="submit" value="Submit">

</form>
</body>
</html>
```

Wlcm.html

```
<html>
<body>
<center>
<h1>Welcome to my website</h1>
</center>
</body>
</html>
```

4 => Java Servlet Programs for Form Handling and Session Tracking

Program:

Index.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Servlet Login Example</title>
</head>
<body>

<h1>Welcome to Login App by Cookie</h1>

<a href="login.html">Login</a> |
<a href="LogoutServlet">Logout</a>

</body>
</html>
```

Link.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Links</title>
</head>
<body>

    <a href="login.html">Login</a> |
    <a href="LogoutServlet">Logout</a>

    <hr>

    <p>... other content goes here ...</p>

</body>
</html>
```

Login.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Login Page</title>
</head>
<body>

    <form action="LoginServlet" method="post">

        Name: <input type="text" name="name"><br>

        Password: <input type="password" name="password"><br>

        <input type="submit" value="login">
```

```
    </form>

</body>
</html>
```

Web.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<web-app xmlns="http://java.sun.com/xml/ns/javaee"
         version="2.5">

    <servlet>
        <servlet-name>LoginServlet</servlet-name>
        <servlet-class>com.javatpoint.LoginServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>LoginServlet</servlet-name>
        <url-pattern>/LoginServlet</url-pattern>
    </servlet-mapping>

    <servlet>
        <servlet-name>LogoutServlet</servlet-name>
        <servlet-class>com.javatpoint.LogoutServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>LogoutServlet</servlet-name>
        <url-pattern>/LogoutServlet</url-pattern>
    </servlet-mapping>

    <welcome-file-list>
        <welcome-file>index.html</welcome-file>
    </welcome-file-list>

</web-app>
```

LoginServlet.java

```
package com.javatpoint;

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class LoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
    throws ServletException, IOException {

        String name = request.getParameter("name");
        String password = request.getParameter("password");

        if ("admin123".equals(password) && name != null &&
!name.trim().isEmpty()) {

            Cookie ck = new Cookie("name", name);
            ck.setMaxAge(3600);
            response.addCookie(ck);

            response.setContentType("text/html");
            PrintWriter out = response.getWriter();
            out.println("Successfully logged in! Welcome, " + name);
            request.getRequestDispatcher("Link.html").include(request,
response);

        } else {
            response.sendRedirect("login.html");
        }
    }
}
```

LogoutServlet.java

```
package com.javatpoint;

import javax.servlet.http.*;

public class LogoutServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request,
HttpServletResponse response) {

        Cookie ck = new Cookie("name", "");
        ck.setMaxAge(0);
        response.addCookie(ck);

        try {
            response.sendRedirect("login.html");
        } catch (java.io.IOException e) {
            e.printStackTrace();
        }
    }
}
```

5 => Creation of Three-Tier Applications Using JSP and Databases

Program:

index.html

```
<html>
<head>
<title>Online Examination</title>
<style>
body {text-align:center; font-family:Courier; color:blue;}
</style>
</head>
```

```
<body>
<h2>ONLINE EXAMINATION</h2>
<hr>
<form action="exam.jsp" method="get">
Roll No: <input type="text" name="rno" required><br><br>
Name: <input type="text" name="name" required><br><br>
<input type="submit" value="Start Exam">
</form>
</body>
</html>
```

exam.jsp

```
<%@ page contentType="text/html; charset=UTF-8" %>
<html>
<head>
<title>Exam Page</title>
<style>
body {color:blue; font-family:Courier; text-align:center;}
form {display:inline-block; text-align:left;}
</style>
</head>
<body>

<h2>ONLINE EXAMINATION</h2>
<hr>

<%
String name = request.getParameter("name");
String rno = request.getParameter("rno");

if(name == null || rno == null || name.isEmpty() || rno.isEmpty()) {
    out.println("Please enter your details first!");
} else {
%>

<form action="result.jsp" method="post">
```

```
<input type="hidden" name="name" value="<%=name%>">
<input type="hidden" name="rno" value="<%=rno%>">

<b>1. What is the capital of India?</b><br>
<input type="radio" name="q1" value="Delhi">Delhi<br>
<input type="radio" name="q1" value="Mumbai">Mumbai<br>
<input type="radio" name="q1" value="Chennai">Chennai<br><br>

<b>2. Which language is used for Android apps?</b><br>
<input type="radio" name="q2" value="Java">Java<br>
<input type="radio" name="q2" value="Python">Python<br>
<input type="radio" name="q2" value="C">C<br><br>

<b>3. Which protocol is used for web communication?</b><br>
<input type="radio" name="q3" value="HTTP">HTTP<br>
<input type="radio" name="q3" value="FTP">FTP<br>
<input type="radio" name="q3" value="SMTP">SMTP<br><br>

<input type="submit" value="Submit Answers">

</form>

<%
}
%>

</body>
</html>
```

result.jsp

```
<%@ page import="java.sql.*" %>
<%@ page contentType="text/html; charset=UTF-8" %>
<html>
<head>
<title>Exam Result</title>
<style>
body {font-family:Courier; color:blue; text-align:center;}
```

```
</style>
</head>
<body>

<h2>EXAM RESULT</h2>
<hr>

<%
try {
    String rno = request.getParameter("rno");
    String name = request.getParameter("name");
    String q1 = request.getParameter("q1");
    String q2 = request.getParameter("q2");
    String q3 = request.getParameter("q3");

    if(rno == null || name == null || q1 == null || q2 == null || q3
== null) {
        out.println("Error: Some data is missing!");
    } else {
        int score = 0;
        if("Delhi".equals(q1)) score++;
        if("Java".equals(q2)) score++;
        if("HTTP".equals(q3)) score++;

        Class.forName("org.sqlite.JDBC");
        Connection con =
DriverManager.getConnection("jdbc:sqlite:C:/sqlite/examDB.db");

        PreparedStatement ps = con.prepareStatement(
            "INSERT INTO student_exam VALUES (?, ?, ?)");
        ps.setInt(1, Integer.parseInt(rno));
        ps.setString(2, name);
        ps.setInt(3, score);
        ps.executeUpdate();

        ps.close();
        con.close();

        out.println("Thank you, <b>" + name + "</b>!");
        out.println("<p>Roll No: " + rno + "</p>");
```

```
        out.println("<p>Score: " + score + " / 3</p>");
        out.println("<a href='index.html'>Take Exam Again</a>");
    }
} catch(Exception e) {
    out.println("Error: " + e.getMessage());
}
%>

</body>
</html>
```

web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee" version="3.1">

    <welcome-file-list>
        <welcome-file>index.html</welcome-file>
    </welcome-file-list>

</web-app>
```

6 => Programs using XML schema and XSL/XSLTS

Program:

CDCatalog.xml

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<?xml-stylesheet type="text/xsl" href="CDCatalog.xsl"?>

<catalog>
    <cd>
        <title>Empire Burlesque</title>
        <artist>Bob Dylan</artist>
        <country>USA</country>
        <company>Columbia</company>
        <price>10.90</price>
        <year>1985</year>
    </cd>

    <cd>
        <title>Hide your heart</title>
        <artist>Bonnie Tyler</artist>
        <country>UK</country>
        <company>CBS Records</company>
        <price>9.90</price>
        <year>1988</year>
    </cd>

    <cd>
        <title>Greatest Hits</title>
        <artist>Dolly Parton</artist>
        <country>USA</country>
        <company>RCA</company>
        <price>9.90</price>
        <year>1982</year>
    </cd>
</catalog>
```

CDCatalog.xsl

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:template match="/">
<html>
<body>
<h2>My CD Collection</h2>

<table border="1">
<tr bgcolor="#9acd32">
<th>Title</th>
<th>Artist</th>
</tr>

<xsl:for-each select="catalog/cd">
<xsl:sort select="artist"/>
<tr>
<td><xsl:value-of select="title"/></td>
<td><xsl:value-of select="artist"/></td>
</tr>
</xsl:for-each>

</table>
</body>
</html>
</xsl:template>

</xsl:stylesheet>
```